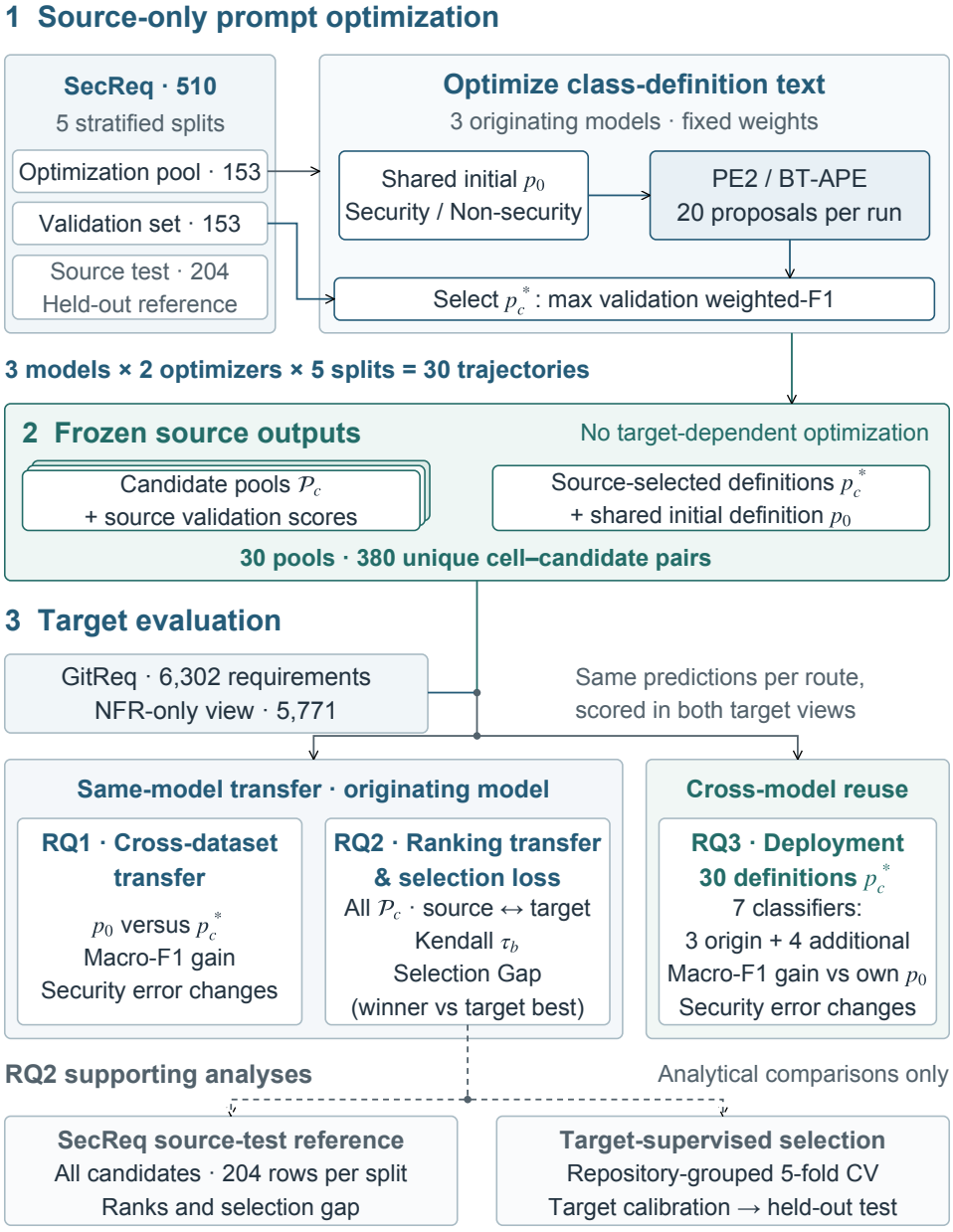

1 Source-only prompt optimization
SecReq · 510
5 stratified splits
Optimization pool · 153
Validation set · 153
Source test · 204
Held-out reference
Optimize class-definition text
3 originating models · fixed weights
Shared initial p_0
Security / Non-security
PE2 / BT-APE
20 proposals per run
Select p_c^* : max validation weighted-F1
3 models × 2 optimizers × 5 splits = 30 trajectories
2 Frozen source outputs
No target-dependent optimization
Candidate pools $\mathcal{P}_c$
+ source validation scores
Source-selected definitions p_c^*
+ shared initial definition p_0
30 pools · 380 unique cell–candidate pairs
3 Target evaluation
GitReq · 6,302 requirements
NFR-only view · 5,771
Same predictions per route,
scored in both target views
Same-model transfer · originating model
RQ1 · Cross-dataset transfer
p_0 versus p_c^*
Macro-F1 gain
Security error changes
RQ2 · Ranking transfer & selection loss
All $\mathcal{P}_c$ · source ↔ target
Kendall τ_b
Selection Gap
(winner vs target best)
Cross-model reuse
RQ3 · Deployment
30 definitions p_c^*
7 classifiers:
3 origin + 4 additional
Macro-F1 gain vs own p_0
Security error changes
RQ2 supporting analyses
Analytical comparisons only
SecReq source-test reference
All candidates · 204 rows per split
Ranks and selection gap
Target-supervised selection
Repository-grouped 5-fold CV
Target calibration → held-out test